

Intro to Wireshark & Bluetooth

October 28th, 2025

CS 151.1001

UNR CSE Department



Bluetooth

Bluetooth

- Wireless protocol
- Uses ISM radio bands
- ISM– are portions of the radio spectrum reserved internationally for industrial, scientific, and medical (ISM) purposes.
- Two types:
 - Low Energy (BLE)
 - Classic



BLE vs. Classic

BLE:

- Data rate: 1Mbps
- Small amounts of data
 - Ex: Fitbits
- Low power consumption
- BLE now serves all of the functions that Bluetooth Classic was designed for.

Classic:

- Data rate: 3Mbps
- Larger amounts of data
 - Ex: Video/audio streaming
- Larger packet size



Almost all mobile devices support both (dual mode).

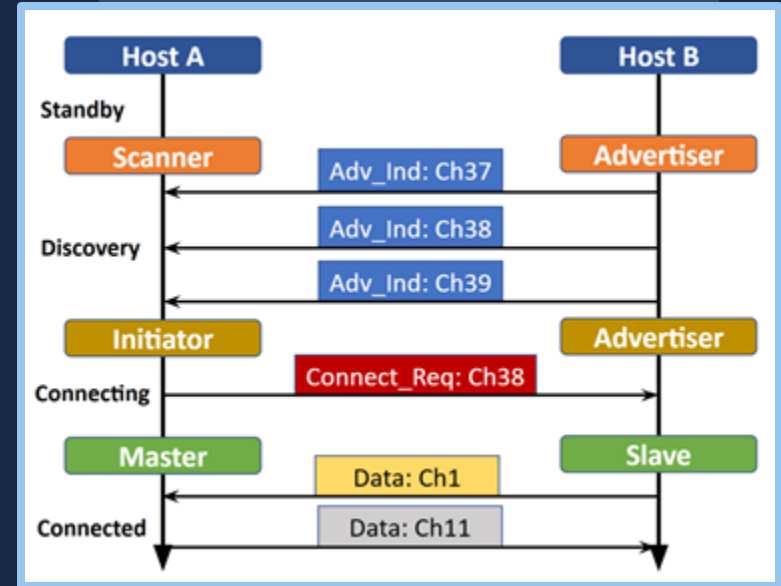
Bluetooth

- Bluetooth defines two transmissions types: data and advertising transmissions.
- All devices initially begin in advertising mode.
- Advertising
 - Device discovery
 - Connection establishment
- Data
 - Bidirectional connection between devices
- Bluetooth LE packet format:

1 (byte)	4 (bytes)	2-257 (bytes)	3 (bytes)
Preamble	Access address	PDU	CRC

Bluetooth

- Initially devices are in Standby state.
- Devices enter a Discovery state where the device wishing to be discovered becomes the Advertiser and host wishing to connect becomes a Scanner.
- The Advertiser sends advertising packets containing basic information about the host.
- All Scanners receive these packets.



Bluetooth Attacks

- Bluejacking– sending unsolicited messages to a Bluetooth-enabled device
- Bluesnarfing– extracting and stealing stored data.
 - Hackers pair to a victim's device by skipping the code pairing process.
- Bluetooth v4 and v5 have encryption now.

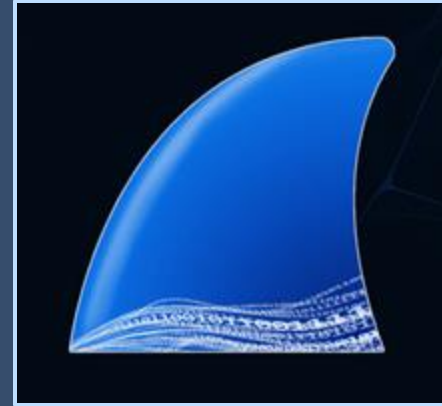




Wireshark

Wireshark

- “Wireshark is a powerful, open-source network protocol analyzer that allows users to capture and interactively browse the traffic running on a computer network, providing deep inspection of hundreds of protocols.” - <https://www.wireshark.org/>
- **Note: DO NOT CAPTURE NETWORK TRAFFIC ON PUBLIC/SHARED WI-FI



Wireshark Steps

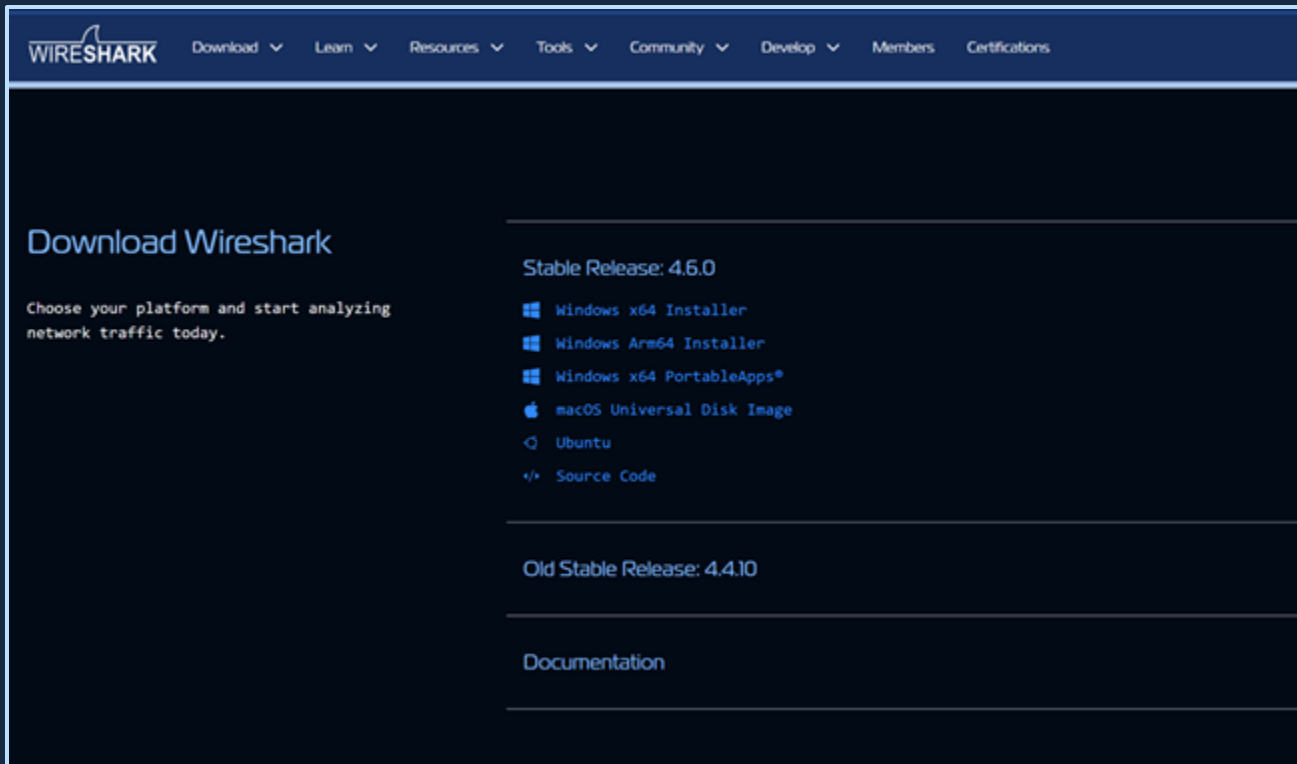
Download & install Wireshark [01]

Open a packet capture file [02]

File > Open > File name > Open [02a]

Analyze and filter packets! [03]

[01]



The screenshot shows the Wireshark website's download page. At the top is a dark blue navigation bar with the Wireshark logo and several menu items: Download, Learn, Resources, Tools, Community, Develop, Members, and Certifications. The main content area has a light blue header with the title 'Download Wireshark'. Below this, a paragraph encourages users to choose their platform and start analyzing network traffic. To the right, there are three sections: 'Stable Release: 4.6.0' with a list of download links for Windows (x64 and ARM64 installers, x64 PortableApps), macOS (Universal Disk Image), Ubuntu, and Source Code; 'Old Stable Release: 4.4.10'; and 'Documentation'.

WIRESHARK

Download ▾ Learn ▾ Resources ▾ Tools ▾ Community ▾ Develop ▾ Members Certifications

Download Wireshark

Choose your platform and start analyzing network traffic today.

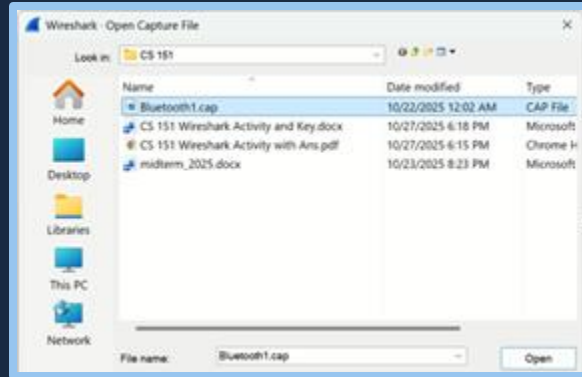
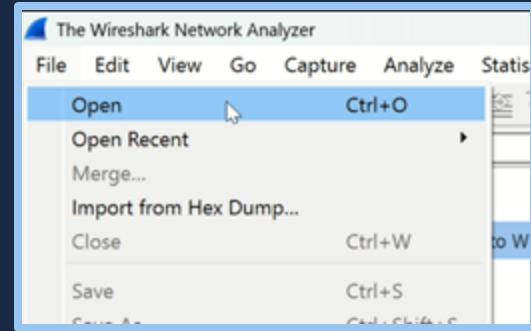
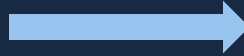
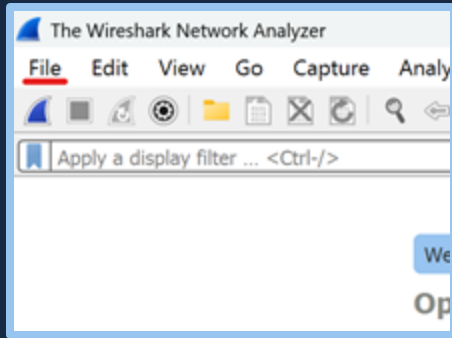
Stable Release: 4.6.0

-  [Windows x64 Installer](#)
-  [Windows Arm64 Installer](#)
-  [Windows x64 PortableApps®](#)
-  [macOS Universal Disk Image](#)
-  [Ubuntu](#)
-  [Source Code](#)

Old Stable Release: 4.4.10

Documentation

[02]



[03]

Bluetooth Log

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter: *Ctrl+L

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	host	controller	HCI_CMD	9	Sent Inquiry
2	0.009559	controller	host	HCI_EVT	7	Rcvd Command Status (Inquiry)
3	2.792499	controller	host	HCI_EVT	18	Rcvd Inquiry Result
4	20.584540	controller	host	HCI_EVT	4	Rcvd Inquiry Complete
5	20.584967	host	controller	HCI_CMD	14	Sent Remote Name Request
6	20.58538	controller	host	HCI_EVT	7	Rcvd Command Status (Remote Name Request)
7	41.024117	controller	host	HCI_EVT	258	Rcvd Remote Name Request Complete
8	82.957128	controller	host	HCI_EVT	13	Rcvd Connect Request
9	82.957154	host	controller	HCI_CMD	11	Sent Accept Connection Request
10	82.972125	controller	host	HCI_EVT	7	Rcvd Command Status (Accept Connection Request)
11	83.029118	controller	host	HCI_EVT	9	Rcvd PIN Code Request
12	83.029392	host	controller	HCI_CMD	27	Sent PIN Code Request Reply
13	83.042135	controller	host	HCI_EVT	13	Rcvd Command Complete (PIN Code Request Reply)
14	83.362862	controller	host	HCI_EVT	14	Rcvd Connect Complete
15	97.074780	controller	host	HCI_EVT	13	Rcvd Connect Request
16	97.074807	host	controller	HCI_CMD	11	Sent Accept Connection Request
17	97.089772	controller	host	HCI_EVT	7	Rcvd Command Status (Accept Connection Request)
18	97.148771	controller	host	HCI_EVT	9	Rcvd PIN Code Request
19	97.149818	host	controller	HCI_CMD	27	Sent PIN Code Request Reply
20	97.162769	controller	host	HCI_EVT	13	Rcvd Command Complete (PIN Code Request Reply)
21	97.378726	controller	host	HCI_EVT	26	Rcvd Link Key Notification
22	97.570691	controller	host	HCI_EVT	14	Rcvd Connect Complete
23	97.570718	host	controller	HCI_CMD	8	Sent Write Link Policy Settings
24	97.571692	controller	host	HCI_EVT	10	Rcvd Page Scan Repetition Mode Change
25	97.585687	controller	host	HCI_EVT	9	Rcvd Command Complete (Write Link Policy Settings)
26	97.585702	host	controller	HCI_CMD	8	Sent Change Connection Packet Type
27	07.000000	controller	host	HCI_EVT	0	Read Max Tx/Rx Flow

Frame 1: 9 bytes on wire (72 bits), 9 bytes captured (72 bits)

- Bluetooth
- Bluetooth HCI N4
- Bluetooth HCI Command - Inquiry

0000 01 01 04 05 33 8b 0e 10 00

Wireshark Activity

[pssstt check WebCampus]

Destination	Protocol	Length	Info
158.194.196	QUIC	231	Payload (Encrypted), PKN: 16, CID: 1155817
168.38.104	QUIC	62	Payload (Encrypted), PKN: 20
168.38.104	QUIC	189	Payload (Encrypted), PKN: 21
168.38.104	QUIC	61	Payload (Encrypted), PKN: 22
158.194.196	QUIC	70	Payload (Encrypted), PKN: 17, CID: 1155817
168.38.104	QUIC	68	Payload (Encrypted), PKN: 23
158.194.196	QUIC	70	Payload (Encrypted), PKN: 18, CID: 1155817
217.1.57	TCP	66	63639 → 80 [SYN] Seq=0 Win=62727 Len=0 MSS=
217.1.57	HTTP	477	GET /flag HTTP/1.1
168.38.104	TCP	66	80 → 63639 [SYN, ACK] Seq=0 Ack=1 Win=6272
217.1.57	TCP	54	63639 → 80 [ACK] Seq=1 Ack=1 Win=262656 Len=0
168.38.104	TCP	54	80 → 63637 [ACK] Seq=1 Ack=424 Win=62336 Len=0
168.38.104	TCP	71	80 → 63637 [PSH, ACK] Seq=1 Ack=424 Win=62
168.38.104	HTTP	263	HTTP/1.0 200 OK (text/html)
217.1.57	TCP	54	63637 → 80 [ACK] Seq=424 Ack=228 Win=26248
217.1.57	TCP	54	63637 → 80 [FIN, ACK] Seq=424 Ack=228 Win=
168.38.104	TCP	54	80 → 63637 [ACK] Seq=228 Ack=425 Win=62336
168.38.104	TCP	54	9000 → 58856 [ACK] Seq=1 Ack=1 Win=1530 Len=0

